



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

Subject: Programming with Java

ASSIGNMENT No :2

TITLE :Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Assignment 2: Methods and Static Members

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Theory

Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values, and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles.

Outcome

Conclusion

We have learnt what is the function of a constructor, the various types of constructors and their working via two different programs

Exercise

1. Create a **Student class** using default and parameterized constructors to initialize student name and roll number.
2. Develop a **Mobile Phone Inventory System** using different constructors to initialize mobile details and create duplicate object records.

Output

Problem1

[Done] exited with code=0 in 0.376 seconds

```
[Running] cd "/Users/raghav_oze/Desktop/java/" && javac Student.java && java Student
Student 1
Name : null
Roll No : 0
```

```
Student 2
Name : agam oze
Roll No : 21
```

[Done] exited with code=0 in 0.376 seconds

Problem2

[Done] exited with code=0 in 0.364 seconds

```
[Running] cd "/Users/raghav_oze/Desktop/java/exp2/" && javac Mobile.java && java Mobile
Mobile 1
Brand : Nothing
Model : Unknown
Price : ₹0
```

```
Mobile 2
Brand : Nothing
Model : 3a Lite
Price : ₹19000
```

```
Duplicate Mobile Record
Brand : Nothing
Model : 3a Lite
Price : ₹19000
```

[Done] exited with code=0 in 0.342 seconds

Code

J Main.java U

◆ .gitattributes

J Student.java U ×

J Student.java >  Student

```
1  class Student {
2      String name;
3      int rollNo;
4
5      // Default
6      Student() {
7          name = "null";
8          rollNo = 0;
9      }
10
11     // Parameterized
12     Student(String n, int r) {
13         name = n;
14         rollNo = r;
15     }
16
17     void display() {
18         System.out.println("Name : " + name);
19         System.out.println("Roll No : " + rollNo);
20     }
21
22     Run main | Debug main
23     public static void main(String[] args) {
24
25         // Using Default Constructor
26         Student s1 = new Student();
27
28         // Using Parameterized Constructor
29         Student s2 = new Student("agam oze", 025);
30
31         System.out.println("Student 1");
32         s1.display();
33
34         System.out.println();
35
36         System.out.println("Student 2");
37         s2.display();
38     }
```

J Mobile.java > Mobile > Mobile()

```
1  class Mobile {
2
3      String brand;
4      String model;
5      int price;
6
7      // Default Constructor
8      Mobile() {
9          brand = "Nothing";
10         model = "Unknown";
11         price = 0;
12     }
13
14     // Parameterized Constructor
15     Mobile(String b, String m, int p) {
16         brand = b;
17         model = m;
18         price = p;
19     }
20
21     // Copy Constructor
22     Mobile(Mobile m) {
23         brand = m.brand;
24         model = m.model;
25         price = m.price;
26     }
27
28     void display() {
29         System.out.println("Brand : " + brand);
30         System.out.println("Model : " + model);
31         System.out.println("Price : ₹" + price);
32     }
33
```

Run main | Debug main

Welcome

Mobile.java ×

Mobile.java > Mobile > Mobile()

1 class Mobile {

33

Run main | Debug main

34 public static void main(String[] args) {

35

36 // Default Constructor

37 Mobile m1 = new Mobile();

38

39 // Parameterized Constructor

40 Mobile m2 = new Mobile("Nothing", "3a Lite ", 19000);

41

42 // Copy Constructor (Duplicate Record)

43 Mobile m3 = new Mobile(m2);

44

45 System.out.println("Mobile 1");

46 m1.display();

47

48 System.out.println();

49

50 System.out.println("Mobile 2");

51 m2.display();

52

53 System.out.println();

54

55 System.out.println("Duplicate Mobile Record");

56 m3.display();

57

58 }